

Instructor run sheet

Reproducible Writing with Quarto

Purpose

This is the facilitator-facing companion to the participant book. The chapters now deliberately contain more explanation and demonstration material than most groups will cover live. Select the examples that suit the audience and revise the schedule below before delivery; the cues, checkpoints, and recovery routes can still be used independently of the timings.

Before the day

- Replace the values in `_variables.yml`.
- Render the book and both participant projects from a fresh directory.
- Open every download archive and test the preflight.
- Confirm the room can access the static book and a backup USB/local copy.
- Bring pre-rendered manuscript, tutorial, slide, and Elsevier outputs.
- Ask participants to complete the prework at least two working days beforehand.
- Arrange a helper if the room has more than about 20 participants.
- Do not update Quarto, R packages, or extensions on the morning of delivery.

Adaptable delivery outline

00:00–00:10 — Setup checkpoint

Say: “Today is about the source-to-output loop. Installation is prework; our first goal is a successful render.”

1. Confirm everyone extracted the starter zip.
2. Run `Rscript preflight.R`.
3. Render `manuscript.qmd` to HTML.

4. Pair anyone with installation trouble and move the room on at minute 10.

Checkpoint: Most participants can see `manuscript.html`.

Fallback: HTML-only participants continue normally. Give anyone with a blocked machine the completed project and pair them for editing.

00:10–00:25 — Reproducible writing

- Draw the source → render → output model.
- Contrast calculation with interpretation.
- Show the project tree and relative CSV path.
- Run the six-minute project-map exercise.
- Take two examples of copy-and-paste risk from the room.

Misconception to surface: Quarto does not make a flawed analysis correct; it makes the recorded instructions rerunnable and reviewable.

00:25–00:50 — Markdown

- Demonstrate source and rendered output side by side.
- Cover YAML, headings, lists, links, equations, callouts, and citations.
- Give 15 minutes for the activity and 5 minutes for repair/review.

Checkpoint: One participant explains why semantic headings are better than manually enlarged text.

00:50–01:25 — Computation

- Explain execution order and clean-session rendering.
- Live-code one cell option, one inline value, and one cross-reference.
- Give 25 minutes for the main activity.
- At minute 80, ask everyone to restart R and render.

Watch for: running lines interactively without saving them into the source; objects inherited from the Global Environment; missing blank lines around fences; duplicate cell labels.

01:25–01:40 — Break

During the break, help PDF installations only if the likely fix takes under five minutes. Otherwise mark participants for the HTML fallback.

01:40–02:10 — Research publishing

- Render HTML and PDF from the same source.
- Ask pairs to identify format-specific differences.
- Demonstrate the bundled Elsevier format from `journal-demo.qmd`.
- Emphasise that journal instructions remain authoritative.

Fallback: Use the pre-rendered Elsevier PDF if local LaTeX packages fail. Do not spend participant activity time debugging a publisher template.

02:10–02:40 — Teaching materials

- Explain the one-source/two-audience problem.
- Show one `unilur-solution` cell and one solution div.
- Give 22 minutes for writing and rendering.
- Search the student HTML for the distinctive solution phrase.

Checkpoint: The phrase occurs in the solution file and not the student file.

02:40–02:52 — Lecture slides

- Preview `slides.qmd`.
- Show slide boundaries, incremental lists, speaker notes, and a computed figure.
- Keep this as a demonstration; do not begin a new participant build.

02:52–03:00 — Capstone

- Ask everyone to restart R and render the manuscript and tutorial.
- Use the capstone checklist.
- Pair-explain how the inline sample size stays synchronised.
- Finish with one concrete next document participants will convert.

Triage rules

1. Preserve the shared pace once the setup block ends.
2. Prefer the completed source over live debugging after five minutes.
3. Treat PDF as a format comparison; HTML is the continuity fallback.
4. Never ask participants to reinstall the entire toolchain during the workshop.
5. Capture recurring errors for a later revision of the troubleshooting appendix.

After the workshop

- Distribute the static book, starter, and completed archives.
- Send the pre-rendered PDFs for accessibility and continuity.
- Share the post-workshop competency assessment and its no-code starter task.
- Remind participants to submit only a safe, de-identified project excerpt or use the supplied task; executable code, data, and R are optional.
- Publish the submission date, support contact, and resubmission process using local learning-and-development procedures.
- Ask one evaluation question tied to the outcome: “Could you render the manuscript from a fresh R session without manual cell execution?”
- Record the tested Quarto and R versions beside the workshop date.